

Fahim F. Chowdhury

Ph.D. Spintronics and quantum computing | Thin-films Module Intern (Intel Corp) | Quantum computing Intern (Lawrence Berkeley National Lab)

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Summary

Ph.D. engineer with 300 mm Intel semiconductor experience in thin-film and PECVD process development, DOE/SPC, and advanced characterization (Ellipsometry, TEM, SEM, XPS, FTIR). Deep expertise in magnetic and semiconductor device physics, with strong publication record in nanoscale spintronics research. Targeting device and technology development roles spanning MRAM, 3D NAND, DRAM, and next-generation nonvolatile memory.

Skills

Device theory: Semiconductor Device Physics, Magnetic Device Physics, Quantum Device Physics.

Metrology: Ellipsometry, Fourier Transform Infra-red (FTIR) Spectroscopy, Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM), X-ray Photoelectron Spectroscopy (XPS), Optical Profilometry.

Electrical & Physical Testing: Magnetic Force Microscopy (MFM), Nitrogen-Vacancy Center Magnetometry, Magneto-Optical Kerr Effect (MOKE), Vector Network Analyzer (VNA), AWGs, pulse generator, oscilloscopes, 4-Point Probing, Cryogenic Measurements, Nanoindentation.

Semiconductor Process Technologies: Plasma-enhanced Chemical Vapor Deposition (PECVD), Atomic Layer Deposition (ALD), Thin Film Dielectrics, Gate-All-Around (GAA) transistor, Photolithography, Electron-Beam Lithography, Plasma Physics, Electron-Beam Evaporation, Sputter Deposition, Lift-Off, wire bonding, Plasma Physics, Design of Experiments (DOE), Statistical Process Control (SPC).

Software & Simulation Tools: Python, JMP, MATLAB, KLayout, SolidWorks, COMSOL Multiphysics, ANSYS HFSS, MuMax3, QuTiP.

Experience

Intel Corporation – Module Engineering Intern | Hillsboro, OR | May 2026–Aug 2026

- Developed PECVD densified, seam-free (DSF) thin films for dielectric gap-fill on 300 mm wafers, leveraging DOE, SPC, and JMP to identify and optimize critical process parameters.
- Developed and optimized SiN deposition recipes for run-rate improvement while meeting thickness, refractive index, wet-etch, and dry-etch performance requirements.
- Characterized and analyzed thin-film properties using ellipsometry, FTIR, XPS, and gravimetric metrology.
- Conducted vendor-supported coupon studies to evaluate loading effects and optimize sidewall-to-bottom deposition ratio using TEM characterization.
- Performed chamber-to-chamber and tool-to-tool process variation studies across the PECVD tools to assess process uniformity.
- Built and maintained a process database for AI/ML modeling, enabling prediction and optimization of key deposition process parameters.

- Collaborated with cross-functional engineering teams to improve process capability, identify process drivers, and support process optimization.

Lawrence Berkeley National Laboratory – Research Intern | Berkeley, CA | Jun 2025–Dec 2025

- Characterized single and two-qubit gate performance and identified critical calibration parameters.
- Performed qubit calibration and randomized benchmarking on 8-qubit transmon QPUs.

Virginia Commonwealth University – Research Assistant | Richmond, VA | Aug 2020–May 2026

- Developed nanoscale quantum control and sensing technologies.
- Reduced microwave excitation power for qubit control by four orders of magnitude while miniaturizing the control hardware size by 100 times.
- Published in Nature Communications, Physical Review Applied, Nano Letters, and Communications Physics.

Auburn University – Research Assistant | Auburn, AL | Jan 2019–July 2020

- Demonstrated connectivity of dense superconducting flexible micro-strip line cables at cryogenic temperature.
- Performed electromagnetic simulations for signal integrity analysis using ANSYS HFSS.

Education

Ph.D. | Virginia Commonwealth University | Richmond, VA

Mechanical and Nuclear Engineering | 2020 - 2026

M.S. | Tuskegee University | Tuskegee, AL

Electrical Engineering | 2016 - 2018

B.S. | Bangladesh University of Engineering and Technology | Dhaka, Bangladesh

Electrical and Electronic Engineering | 2008 - 2013

Professional Development

Statistical Thinking for Industrial Problem Solving (STIPS)

- Statistical Process Control (SPC), Design of Experiments (DOE), Data Analysis, Statistical Modeling, Regression Analysis, Hypothesis Testing, Machine Learning.

Selected Publications

- **Chowdhury, M.F.F.**, Jung, A., La Spina, L., Bartasyte, A., Margueron, S. and Atulasimha, J., 2026. Coherent quantum control of nitrogen vacancy spin with nanoscale magnets. ***Nature Communications***.
- **Chowdhury, M.F.F.**, Niknam, M., Rajib, M.M., Bouchard, L.S. and Atulasimha, J., 2024. Proximal quantum control of spin and spin ensemble with localized control field from skyrmions. ***Physical Review Applied***, 22(6), p.064077.
- **Chowdhury, M.F.F.**, Misba, W.A., Rajib, M.M., Edwards, A.J., Bhattacharya, D., Varghese, M.S., Friedman, J.S. and Atulasimha, J., 2022. Focused surface acoustic wave induced nano-oscillator based reservoir computing. ***Applied Physics Letters***, 121(10).